MIL-W-22759/1E
10 September 1976
SUPERSEDING
MIL-W-22759/1D
10 July 1974
See also "Supersession Data" on Page 3

MILITARY SPECIFICATION SHEET

WIRE, ELECTRIC, FLUOROPOLYMER-INSULATED, TFE AND TFE-COATED-GLASS, SILVER-COATED COPPER CONDUCTOR, 600-VOLT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the wire described herein shall consist of this document and the issue in effect of Specification MIL-W-22759

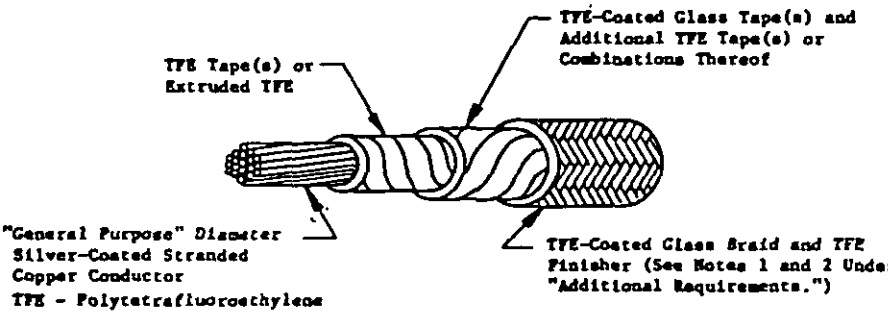


TABLE I. CONSTRUCTION DETAILS

Part No. 1/	Wire size	Stranding (Number of strands X AWG gage of strands)	Diameter of stranded conductor (inches)		Finished wire		
			(min)	(max)	Resistance at 20°C (68°F) (ohms/1000 ft) (max)	Diameter (inches)	Weight (lbs/1000 ft) (max)
M22759/1-22-*	22	19 X 34	.029	.032	15.1	.084 ±.004	7.5
M22759/1-20-*	20	19 X 32	.037	.040	9.19	.094 ±.004	9.5
M22759/1-18-*	18	19 X 30	.046	.050	5.79	.105 ±.005	13.0
M22759/1-16-*	16	19 X 29	.052	.057	4.52	.120 ±.005	16.5
M22759/1-14-*	14	19 X 27	.065	.072	2.88	.138 ±.005	23.0
M22759/1-12-*	12	37 X 28	.084	.089	1.90	.157 ±.005	32.1
M22759/1-10-*	10	37 X 26	.106	.112	1.19	.181 ±.007	47.0
M22759/1-8-*	8	133 X 29	.158	.169	.658	.248 ±.007	82.0
M22759/1-6-*	6	133 X 27	.198	.213	.418	.293 ±.010	122.
M22759/1-4-*	4	133 X 25	.250	.268	.264	.355 ±.015	180.
M22759/1-2-*	2	665 X 30	.320	.340	.170	.420 ±.015	275.
M22759/1-1-*	1	817 X 30	.360	.380	.139	.470 ±.015	348.
M22759/1-01-*	0	1045 X 30	(E) .395	.425	.108	.515 ±.020	429.
M22759/1-02-*	00	1330 X 30	(E) .440	.475	.085	.575 ±.020	542.
M22759/1-03-*	000	1665 X 30	(E) .500	.540	.068	.640 ±.020	668.
M22759/1-04-*	0000	2109 X 30	(E) .563	.605	.054	.710 ±.020	835.

1/ PART NO.: The asterisks in the part number column, Tables I through III, shall be replaced by color code designators in accordance with MIL-STD-681. Examples: Size 20, white - M22759/1-20-9; white with orange stripe - M22759/1-20-93.

(E) denotes changes

FSC 6145

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TABLE II. PERFORMANCE DETAILS

Part No.	Abrasion resistance (Procedure I)					Bend testing				
	Resistance (inches of tape) (min)		Weight support bracket	Weight (lbs)	Tension load (lbs)	Mandrel diameter (inches) (±3%)			Test load (lbs) (±3%)	
	Initial	After immersion 1/				Life cycle (oven & bend tests) 2/	Cold bend test	Wrap test	Life cycle (oven & bend tests) 2/	Cold bend test
M22759/1-22-*	22	15	A	1.0	1.0	.375	.50	.250	.75	2.0
M22759/1-20-*	22	15	A	1.0	1.0	.375	.50	.250	.75	2.0
M22759/1-18-*	22	15	A	1.0	1.0	.500	.75	.375	1.0	2.0
M22759/1-16-*	30	21	A	1.0	2.0	.500	.75	.375	1.0	3.0
M22759/1-14-*	13	9	B	3.0	2.0	.625	1.00	.50	1.0	3.0
M22759/1-12-*	17	12	B	3.0	2.0	.75	2.00	.50	3.0	3.0
M22759/1-10-*	18	13	B	3.0	2.0	1.00	3.00	.75	3.0	5.0
M22759/1-8-*	25	18	B	3.0	2.0	2.00	3.00	1.00	3.0	5.0
M22759/1-6-*	25	18	C	3.0	2.0	3.00	4.00	1.25	6.0	10.0
M22759/1-4-*	33	23	C	4.25	2.0	4.00	6.00	1.50	6.0	10.0
M22759/1-2-*	34	24	C	4.25	2.0	6.50	6.00	2.00	6.0	15.0
M22759/1-1-*	35	25	C	4.25	2.0	10.0	10.0	2.50	6.0	15.0
M22759/1-01-*	48	34	C	4.25	2.0	10.0	10.0	3.00	10.0	20.0
M22759/1-02-*	48	34	C	4.25	2.0	10.0	18.0	4.00	10.0	25.0
M22759/1-03-*	48	34	C	4.25	2.0	10.0	18.0	5.00	10.0	30.0
M22759/1-04-*	48	34	C	4.25	2.0	10.0	18.0	6.00	10.0	30.0

1/ Qualification test only.

2/ Also for bend tests after immersion.

ADDITIONAL REQUIREMENTS

TEMPERATURE RATING: 200°C (392°F) max conductor temperature
VOLTAGE RATING: 600 volts (rms) at sea level
SPARK TEST OF PRIMARY INSULATION: Not required
(E) IMPULSE DIELECTRIC TEST: 6.5 kilovolts (peak), 100% test
INSULATION RESISTANCE: 5000 megohms for 1000 ft (min)
WRAP TEST: Mandrel test required; no cracking
Dielectric test after mandrel wrap, 2500 volts (rms), 60Hz
BLOCKING: 260 ±2°C (500 ±3.6°F)
SHRINKAGE: 0.125 inch max at 200 ±2°C (392 ±3.6°F)
WICKING:
Sizes 22 through 12 - (Procedure II): 2.0% max weight increase, 0.750 inch (max) dye travel
Sizes 10 through 0000: No requirement
LOW TEMPERATURE (COLD BEND):
Bend temperature: -65 ±2°C (-85 ±3.6°F)
Dielectric test, 2500 volts (rms), 60Hz
THERMAL SHOCK: Oven temperature 200 ±2°C (392 ±3.6°F)
Max change in measurement Sizes 22 through 12: 0.060 inch
Sizes 10 through 8: 0.100 inch
Sizes 6 through 0000: 0.125 inch
FLAMMABILITY: Post-flame dielectric test required, 600 volts (rms), 60Hz
LIFE CYCLE: Oven temperature 313 ±2°C (595.4 ±3.6°F)
Dielectric test, 2500 volts (rms), 60 Hz
DIELECTRIC TEST AFTER IMMERSION: 2500 volts (rms), 60Hz
ABRASION RESISTANCE AFTER IMMERSION: Qualification test only
HUMIDITY RESISTANCE: 5000 megohms for 1000 ft, min insulation resistance after humidity exposure

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ADDITIONAL REQUIREMENTS (Continued)

SURFACE RESISTANCE:
Sizes 22 through 12: 50 megohm-inches (min) initial and final readings
Sizes 10 through 0000: No requirement
SMOKE: 313°C (595.4°F)
ACID RESISTANCE: Required
Dielectric test, 2500 volts (rms), 60Hz
COLOR: In accordance with MIL-STD-104, Class 1; white preferred

COLOR STRIPING OR BANDING DURABILITY: 50 cycles (100 strokes) (min), 125 grams weight
IDENTIFICATION DURABILITY: 50 cycles (100 strokes) (min), 125 grams weight
WIRE LENGTH REQUIREMENTS: Schedule B
NOTES (INSULATION):
1. The TFE-coated-glass outer braid shall be tightly applied with no gaps apparent when examined visually without the aid of magnification.
2. The insulation, including the outer braid, shall be heat sintered so that it cannot be easily delaminated.

SUPERSESSION DATA: The wire of this specification sheet, by part number, replaces and supersedes Class 1 wire of MIL-W-7139 (canceled) in accordance with Table III.

TABLE III. SUPERSESSION BY PART NUMBER

Size designation MIL-W-7139 Class 1	Part number MIL-W-22759/1
22	M22759/1-22-*
20	M22759/1-20-*
18	M22759/1-18-*
16	M22759/1-16-*
14	M22759/1-14-*
12	M22759/1-12-*
10	M22759/1-10-*
8	M22759/1-8-*
6	M22759/1-6-*
4	M22759/1-4-*
2	M22759/1-2-*
1	M22759/1-1-*
0	M22759/1-01-*
00	M22759/1-02-*
000	M22759/1-03-*
0000	M22759/1-04-*

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Custodians:
Navy - AS
Army - EL
Air Force - 99

Review activities:
Navy - EC, OS
Army - ME, MI, MU
Air Force -
DSA - IS
NSA

Preparing activity:
Navy - AS
(Project No. 6145-0701-1)

User activities:
Navy - MC
Army - AT, AV
Air Force - 11

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	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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